



The internet bots arms race

There is a secret war going on between scraper bots on the internet, to know the prices of competition. <https://digit.in/sep18-22-1>



Trump filters to fix the net

The "Elders of the Internet" have apologised for social media, and recommend Trump filters to fix the internet. <https://digit.in/sep18-22-2>

The many inventions of the mouse

The humble computer mouse was invented at least three times!

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he very first mice, were actually built for analog computers - and not digital mouse. An easy to use input

device was required for RADAR and SONAR systems in warships, which lead to the invention of "tracking balls". Think of them as inverted, exposed mice, where the user directly manipulates the ball, instead of rolling it along a surface.

THE SECRET MICE

The first mouse was developed for an analogue computer of the UK's Royal Navy. It was soon after the end of the second world war, and Professor Ralph Benjamin was working on the Comprehensive Display System, a system to calculate the trajectories of planes. The cursor on the system was controlled by a joystick. It was this version of the system that went into production. However, a single version of the system had a trackball for manipulating the cursors. Within a metal casing, was a metal ball, which was in contact with two rubber balls. Each of the rubber balls were for providing directional input. The metal ball itself was the only exposed part, which was directly manipulated by the user. The user cupped their hands over the metal ball, very similar to the way we use mice today. The "ball



The DATAR Trackball

tracker" was patented in 1946, but the British Admiralty kept the invention a secret. The versions of the system actually installed in ships had a joystick instead of the ball tracker. The Navy was not interested in commercially exploiting the invention. Because of its status as a military secret, Benjamin did not receive widespread credit for the invention, and as such is not widely recognised as the inventor of the computer mouse.

In 1949, The Royal Canadian Navy started working on its own tracking system for warships. The Digital Automated Tracking and Resolving (DATAR) system was made in collaboration with Ferranti Canada. Tom Cranston, Fred Longstaff and Kenyon Taylor, engineers at Ferranti Canada came up with their own version of the trackball. The device used four wheels arranged around a ball, which could be directly manipulated to move a cursor on the screen. Because of the extreme budget constraints the engi-

neers could not even make a trackball from scratch, and ingeniously used an off the shelf bowling ball. Instead of two rubber balls in the UK Navy's version, the Canadian Navy's version of the trackball used metal wheels. The trackball was installed in the front of the command console. The input device for the DATAR system was also kept a military secret.

THE MODERN MOUSE

On 9 December, 1968, Douglas Engelbart and a group of researchers demonstrated the capabilities of the oN-Line System or NLS. There were around 1,000 technology professionals in attendance at the Fall Joint Computer Conference in San Francisco. The demonstration, among other things, showed the computer mouse. A cursor was used to manipulate the text in a word document. The ghosting cursor left behind a short trail as it travelled across the screen. It is because of this demonstration, that Engelbart is